Quantum mechanics – waves, probability and measurement

Quantum mechanics is the most precisely tested theory humanity has produced. It explains why atoms are stable, why materials have colour, how transistors and lasers work, and why the Sun shines. Yet there is still disagreement about what it means. This page covers the structure of the theory – from Planck's quanta via Schrödinger's equation to entanglement and Bell's inequality – with two interactive figures in which you can solve the Schrödinger equation yourself and watch a wave packet tunnel.

The old quantum theory (1900–1924)

On 14 December 1900 Max Planck presented a formula for black-body radiation that fitted the measurements – but only if energy was exchanged in portions $E = h\nu$. He called it "an act of desperation". In 1905 Einstein went further: light itself consists of quanta, photons, and that explains the photoelectric effect, in which the energy of the electrons depends on the frequency of the light, not on its intensity. Robert Millikan tried for ten years to disprove it – and confirmed it in 1916.

In 1913 Niels Bohr explained the spectrum of hydrogen with electrons in definite orbits with angular momentum $n\hbar$, which emit light only when they jump between orbits. Arnold Sommerfeld extended the model to elliptical orbits. In 1914 James Franck and Gustav Hertz showed, with electrons colliding with mercury atoms, that atoms can take up only certain energies. But the "old quantum theory" was a patchwork of classical mechanics and ad hoc rules, and it failed for helium and for the intensities of spectral lines.

Matter as waves

In his doctoral thesis of 1924 Louis de Broglie proposed that if light can behave like particles, particles can also behave like waves, with wavelength

$$\lambda = \frac{h}{p}$$

In 1927 Clinton Davisson and Lester Germer at Bell Labs and George Paget Thomson in Aberdeen saw electrons produce interference patterns from crystals – exactly like X-rays. (G.P. Thomson received the Nobel Prize for showing that the electron is a wave; his father J.J. Thomson had

received it for showing that it is a particle.) In 1989 Akira Tonomura at Hitachi performed the double-slit experiment with single electrons: each electron hits the screen in one place, but thousands of electrons slowly build up the interference pattern. See Schrödinger's cat – an information paradigm.

Schrödinger's equation

Two versions of the new mechanics appeared almost at the same time. In June 1925, on the island of Heligoland where he had gone to cure his hay fever, Werner Heisenberg found a mechanics in which only measurable quantities – frequencies and intensities – appeared. Max Born and Pascual Jordan realised that Heisenberg's tables were matrices that do not commute. Over the Christmas holiday of 1925–26 in Arosa Erwin Schrödinger found a wave equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi = -\frac{\hbar^2}{2m}\nabla^2\Psi + V(\vec{r})\Psi$$

For stationary states with definite energy it becomes the eigenvalue problem $\hat{H}\psi = E\psi$. Schrödinger soon showed that the two theories are mathematically equivalent. Paul Dirac and John von Neumann gave the theory its abstract form with state vectors in a Hilbert space. In 1928 Dirac united quantum mechanics with special relativity and predicted antimatter.

The postulates of quantum mechanics

1. States

The state of a system is described by a normalised vector $|\psi\rangle$ in a Hilbert space – for a particle, a wave function $\psi(\vec{r})$. Superpositions of allowed states are also allowed.

2. Observables

Each measurable quantity corresponds to a Hermitian operator $\hat{A}$. The possible results of a measurement are its eigenvalues a_n. Position and momentum: $[\hat{x}, \hat{p}] = i\hbar$.

3. Born's rule

The probability of measuring a_n is $|\langle a_n|\psi\rangle|^2$. For a particle, $|\psi(\vec{r})|^2$ is the probability density for finding it at $\vec{r}$ (Max Born, 1926).

4. Evolution and measurement

Between measurements the state evolves deterministically according to Schrödinger's equation. After a measurement with result a_n the system is in the state $|a_n\rangle$.

Postulate 4 is the one that creates the problems of interpretation: what is a "measurement", and why does it not follow Schrödinger's equation?

The uncertainty relation and complementarity

In 1927 Heisenberg showed that operators that do not commute cannot have sharp values at the same time:

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}, \qquad \Delta E\,\Delta t \gtrsim \frac{\hbar}{2}$$

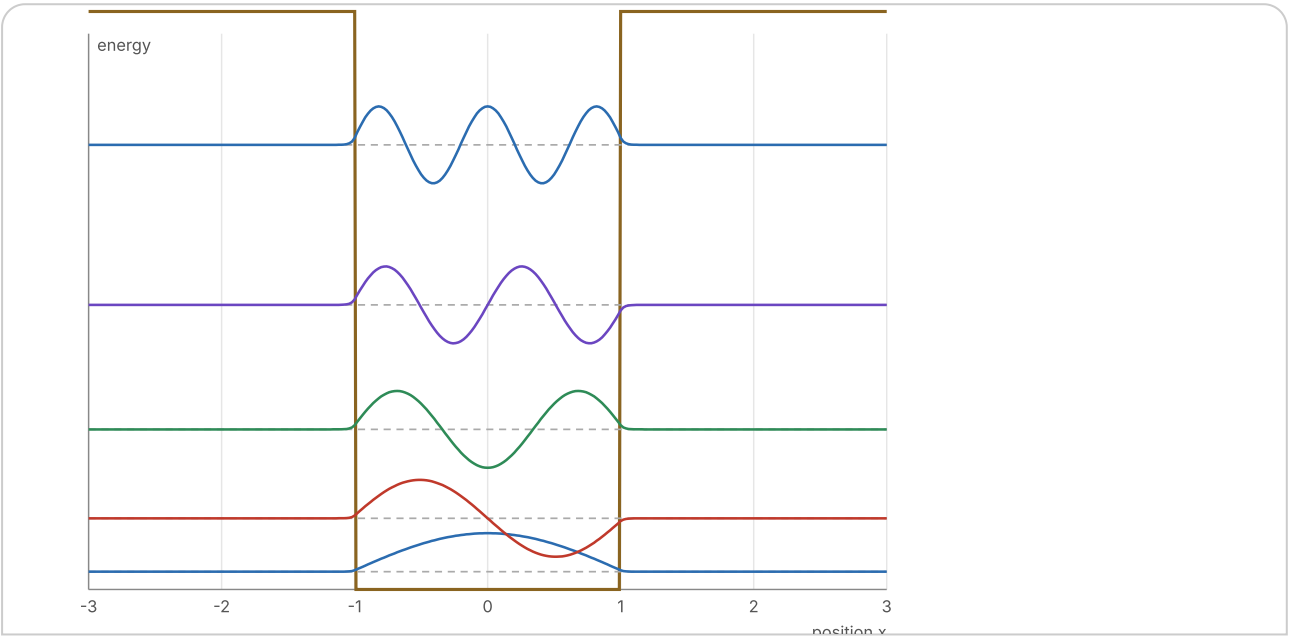
This is not a limitation of the measuring equipment but a property of the states themselves: a wave packet that is narrow in space is necessarily broad in momentum (just as a short pulse of sound contains many frequencies). The same year Bohr formulated the principle of complementarity: the wave and particle pictures exclude each other in any single experiment, but both are needed for a complete description. This was the core of the so-called Copenhagen interpretation, debated at the Solvay conference in Brussels in October 1927.

Bound states

A particle trapped in a potential well can have only certain energies – the quantisation that Bohr had to postulate, but which here follows from the requirement that the wave function must go to zero far away. Some classic results:

Potential	Energy levels
Infinite well of width a	$E_n = \dfrac{n^2\pi^2\hbar^2}{2ma^2},\ n = 1, 2, 3, \ldots$
Harmonic oscillator	$E_n = \hbar\omega\left(n + \tfrac{1}{2}\right),\ n = 0, 1, 2, \ldots$
Hydrogen atom	$E_n = -\dfrac{13.6\ \text{eV}}{n^2}$

Note that the lowest energy is never zero: the zero-point energy is a direct consequence of the uncertainty relation. It is the reason helium does not freeze at normal pressure.

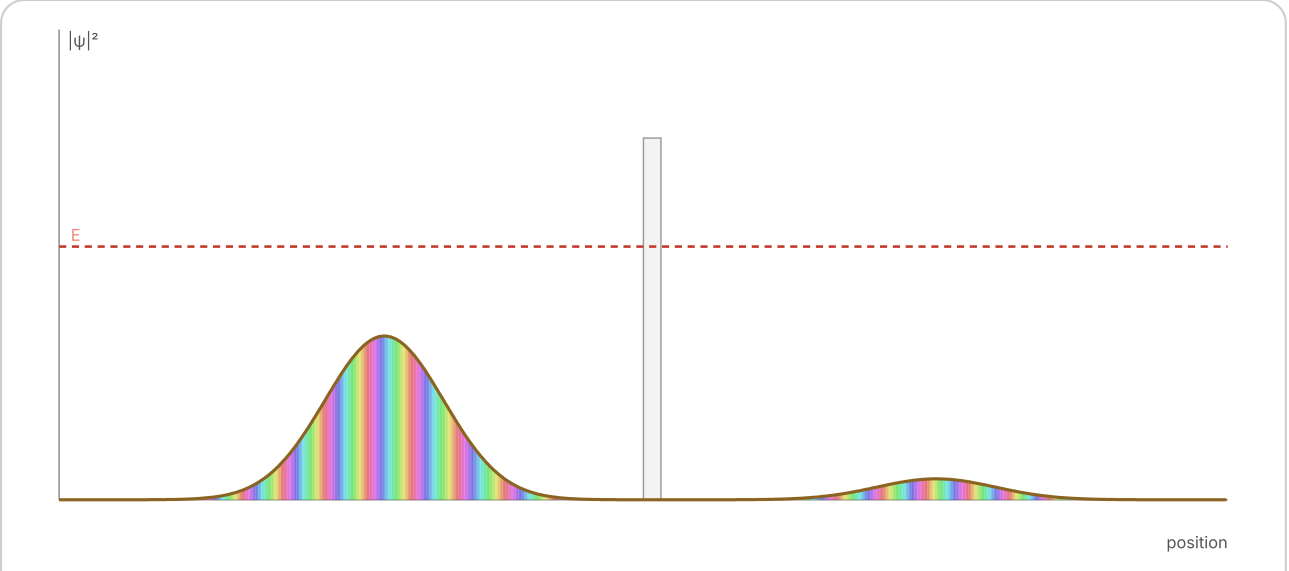


Tunnelling

A classical ball rolling towards a hill higher than its energy rolls back. A quantum particle has a certain probability of getting through. For a rectangular barrier of height V_0 and width a the transmission probability for $E < V_0$ is

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)}\right]^{-1}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

For a thick barrier $T \approx e^{-2\kappa a}$ – exponentially sensitive to the width. In 1928 George Gamow explained alpha decay by tunnelling (see Nuclear physics). Tunnelling makes fusion in the Sun possible, it is used in tunnel diodes and flash memory, and in the scanning tunnelling microscope (Binnig and Rohrer, 1981) the tunnelling current between a tip and a surface is measured so precisely that individual atoms can be seen. In the 1980s Anthony Leggett proposed that even a macroscopic current in a superconducting circuit can tunnel – this was confirmed and is the basis of superconducting qubits.



The hydrogen atom

Schrödinger's first application was the hydrogen atom. With a Coulomb potential the equation separates into a radial part and an angular part, and three quantum numbers emerge: the principal quantum number n, the orbital quantum number $l = 0, \ldots, n-1$ and the magnetic quantum number $m = -l, \ldots, l$. The energy depends only on n and gives exactly Bohr's formula – but now with wave functions, orbitals, instead of orbits. See the interactive pages on hydrogen orbitals and spectra.

Finer details require more: the fine structure (relativistic corrections and spin–orbit coupling, Dirac's equation), the Lamb shift (interaction with vacuum fluctuations, measured by Willis Lamb in 1947 and explained by quantum electrodynamics) and the hyperfine structure (the spin of the nucleus), which gives hydrogen's 21 cm line, the most important in radio astronomy.

Spin

In 1922 in Frankfurt Otto Stern and Walther Gerlach sent a beam of silver atoms through an inhomogeneous magnetic field. Classically the beam should spread out; instead it split in two. In 1925 Samuel Goudsmit and George Uhlenbeck proposed that the electron has an intrinsic angular momentum, *spin*, with only two settings, $\pm\hbar/2$. Pauli described spin with 2×2 matrices, and Dirac showed that spin follows automatically from relativistic quantum mechanics. Spin is the simplest quantum system: a *qubit*. Measure the spin along z and then along x, and the result is completely random; measure along z again, and the first result has been "forgotten". This is uncertainty in its purest form. See Electron spin resonance.

Identical particles

Two electrons cannot be told apart, not even in principle. The wave function of two identical particles must therefore be either symmetric (bosons, integer spin) or antisymmetric (fermions, half-integer spin) under exchange. Antisymmetry gives Pauli's exclusion principle, which explains the periodic table, the stability of matter and white dwarfs. Symmetry gives lasers and Bose–Einstein condensates. See Quantum statistics and the periodic table.

Entanglement and Bell's inequality

In 1935 Einstein, Boris Podolsky and Nathan Rosen (EPR) wrote that quantum mechanics must be incomplete: two particles can be in a state where a measurement on one instantly determines the result on the other, however far apart they are. Schrödinger called the phenomenon *entanglement* (Verschränkung) and the same year wrote about his cat. In 1964 John Bell showed that no local theory with hidden variables can reproduce the predictions of quantum mechanics, and experiments from Clauser (1972) and Aspect (1982) to the "loophole-free" experiments of 2015 have confirmed quantum mechanics. Today entanglement is a resource for quantum cryptography, teleportation and quantum computers. See From Bell's Inequality to Sound Intensity.

What does it mean?

No one disagrees about how to calculate with quantum mechanics. But about what it says about reality there are still several schools:

Copenhagen

Bohr and Heisenberg: the wave function is a tool for predicting the results of measurements. It makes no sense to speak of properties that are not measured.

Many worlds

Hugh Everett (1957): the wave function never collapses. All outcomes happen, each in its own branch of the universe. David Deutsch is one of its strongest advocates.

Pilot wave

de Broglie (1927) and David Bohm (1952): particles have definite positions at all times and are guided by the wave function. The theory is deterministic but non-local.

Collapse theories and decoherence

Ghirardi, Rimini and Weber (1986) propose a real, spontaneous collapse for large systems. Decoherence theory (Zeh, Zurek) explains why superpositions become invisible for large objects, but not by itself why we see a single outcome.

Leggett pointed out that the question of where the boundary between the quantum and the classical world lies can be tested experimentally, and today superpositions have been made of molecules with thousands of atoms and of micromechanical oscillators. See Anthony Leggett and Schrödinger's cat.

Further reading

- David J. Griffiths: *Introduction to Quantum Mechanics* (3rd ed., Cambridge 2018)
- P.A.M. Dirac: *The Principles of Quantum Mechanics* (4th ed., Oxford 1958)
- Richard P. Feynman: *The Feynman Lectures on Physics*, vol. III (1965)
- Alastair I.M. Rae: *Quantum Mechanics* (6th ed. 2015) – a textbook from Birmingham
- J.J. Sakurai and Jim Napolitano: *Modern Quantum Mechanics* (3rd ed. 2020)
- Manjit Kumar: *Quantum. Einstein, Bohr and the Great Debate about the Nature of Reality* (2008)

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